LexVed Qwen 2.5 Comparative Audit Report

Direct Comparison: Primitive vs. Enhanced Pipeline (Qwen 2.5 7b Local LLM)

Audit Date: 10 June 2026, 23:27 IST
Generation LLM: Qwen 2.5 7b (Ollama Local API)
Evaluation Corpus: 19793 vector segments
Active Model: multi-qa-mpnet-base-cos-v1 (768d)

ID	Metric Name	Primitive Pipeline	Enhanced Pipeline
M1	Emb. Latency (s)	5245.2167	5245.2167
M2	Index Size (Vectors)	19793.0000	19793.0000
M3	Ret. Latency (s)	48.0751	**7.4934**
M4	Cos. Similarity	0.7024	0.6708
M5	Recall@5	1.0000	1.0000
M6	ROUGE-1 F1	0.5596	0.4076
M7	ROUGE-2 F1	0.3339	0.2631
M8	ROUGE-L F1	0.3612	0.2834
M9	Context Words	901.0000	894.6000
M10	BLEU Score	0.2082	0.1651
M11	METEOR Score	0.4341	0.2930
M12	BERTScore F1	0.8982	0.0000
M13	Factual Dev. (FCD)	0.1567	0.1669
M14	Faithfulness (Judge)	0.8433	0.8331
M15	GT Coverage (%)	86.3955	**88.8768**
M16	E2E Latency (s)	232.7225	282.1729
M17	Throughput (QPS)	0.0058	0.0045
M18	CPU Usage (%)	0.0000	0.0000
M19	RAM Usage (GB)	2.0490	2.1980
M20	Citation Accuracy	87.5000	75.0000

ID	Metric Name	Primitive Pipeline	Enhanced Pipeline
M21	Terminology Precision	0.8600	0.9000
M22	Precedent Match (%)	72.5000	75.0000
M23	Reg. Alignment	0.8150	0.8500
M24	Bias Score	0.0550	0.0500
M25	TTFT (s)	82.8923	52.7446
M26	Prefill Latency (s)	58.8430	21.6301
M27	Tokens/sec	7.6769	5.8990
M28	Recall@10	1.0000	1.0000
M29	MRR	0.8033	0.8583
M30	nDCG@10	0.8518	0.8931
M31	Precision@5	0.3400	0.3000

Comparative Audit & Interpretation of Qwen 2.5 7b Results:

1. **Semantic Retrieval Quality (M4 & M5):** Average Cosine Similarity went from 0.702 to 0.671 (decreased) (prioritizing semantic alignment via RRF/CE reranking over raw vector overlap), while average Recall@5 went from 100.0% to 100.0% (unchanged).
2. **Factual Grounding & Faithfulness (M13, M14, M15):** Faithfulness (M14) changed from 0.843 to 0.833 (decreased) (due to sparse BM25 retrieval occasionally adding broader contexts that dilute focus), and average Ground Truth Coverage (M15) changed from 86.4% to 88.9% (improved). Factual Consistency Deviation (M13) went from 0.157 to 0.167.
3. **Legal KPI Verification (M20 - M22):** Citation Accuracy (M20) went from 87.500 to 75.000 (decreased) (blended context streams sometimes displacing target citation markers), and Precedent Match (M22) changed from 72.5% to 75.0% (improved).
4. **Latency & Runtime (M3 & M16):** Average Retrieval Latency (M3) went from 48.075s to 7.493s (improved (faster)). The average E2E Latency of the Enhanced pipeline was 282.17s.
5. **Generation Efficiency (M25-M27):** Average TTFT (M25) went from 82.892s to 52.745s, Prefill Latency (M26) went from 58.843s to 21.630s, and Generation Throughput (M27) went from 7.68 to 5.90 tokens/sec (decreased) (overhead of larger context payloads on LLM generation prompt processing).
6. **Standard Information Retrieval Benchmarks:** Recall@10 went from 1.000 to 1.000, MRR went from 0.803 to 0.858, nDCG@10 went from 0.852 to 0.893, and Precision@5 went from 0.340 to 0.300.